# ETHICAL HACKING

From scratch

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# Syllabus

- Introduction to course
- setting up hacking lab
- linux & windows intro
- linux & windows basic commands & scripting
- Git basics
- Networking fundamentals
- Ports & protocols
- Network security & SOC
- Phases of hacking
- cryptography
- Steganography
- Anonymity

# Syllabus

- Wifi networks hacking
- Android & system hacking
- Dos & DDos
- Sniffing & Spoofing
- Password Cracking
- social Engineering

# Introduction to course

- Introduction to Ethical hacking
- Types of hacker
- Terms in Hacking
- Phases of hacking
- Components of computer
- what is OS & Kernel
- What is Shell & imp layers of OS
- How OS starts
- How program works & compiler vs interpreter
- Decompiler vs Disassembler
- Making a system call
- user space vs kernel space

# Setting up hacking lab

- Virtualization tech & hypervisors
- Download vm ware & setup
- Download kali linux & setup in vm
- Download windows 10 and setup
- Download Metasploitable 2 and setup
- VMs login & basic info

# linux & windows intro

- windows vs linux
- File system linux vs windows
- logs,registers & importat software

# Linux & windows basics commands &scrpting

- shells & shorcuts
- linux file handling
- windows file handling
- file editing linux & windows
- basic to advaced commands

# Git basics

- Git basic usage

# Network fundamentals

- What is network & ARPNET
- Networking devices
- Types of networks
- IP,IP types , MAC , internet working &NAT
- IPv4 & Network interfaces
- Subnetting & types
- IP classes subnetting
- IPv6 basics & online subnet calculator

# Ports & Protocols

- Ports
- Ports practical (netstat&ss)
- Protocols, TCP vs UDP
- TCP-IP model & osi Model
- Encapsulation & Decapsulation
- Wireshark intro & practical TCP-IP
- IP packet theory & practical
- TCP 3way handshak, practical view of TCP& UDP
- FTP protocol practical & usage
- SMB practical & usage
- NFS networking configuration
- NFS practical & usge
- SSH practical & configuration usage

# Ports & Protocols

- telnet practical&usage
- HTTP protocol & usage
- HTTPS & its decryption
- DHCP practical & usage
- DNS & its usage
- ICMP & traceroute
- ARP & its usage
- SMTP & practical usage
- SNMP & practical usge

# Networking security & soc

- Firewall practical, windows firewall vs UFW
- IDS-IPS pratical & theory
- SOC
- SIEM & Wazuh practical
- EDR & Velociraptor Practical

# Phases in Hacking

- Foot printing (or) information gathering
- Scanning
- Gaining access
- Maintaining access
- Covering access

# Cryptography

- Cryptography & types of ciphers
- Encoding & decoding
- Encryption , symmetric & assymetric practical
- Hashing

# Steganography

- Steganography
- Video steganography practical
- Audio steganography practical
- Image based steganography
- white space steganography practical

# Anonymity

- webproxy practical
- VPN practical
- TOR practical
- proxychains practical
- darkweb & hidden services

# Wifi networks hacking

- Intro to wifi networks
- WIFI adaptors
- WPA2-PSK attacks (theory)
- Connecting wifi adaptor to vmware
- Changing Mac address of the adaptor
- Enabling & Disabeling monitor mode
- Capturing Handshake
- Deauthentication attack to capture handshake
- Generating wordlist & cracking password
- Evil twin attack
- ARP cache spoofing attack
- DNS cache spoofing attack

# IMPORTANT TOPICS

- Andorid & system hacking
- DOS & DDOS
- pasword cracking
- social engineering

# THANK YOU

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